LAB_ASSIGMENT:2

(2401010021_ANSHIKA)

Objective:

• To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

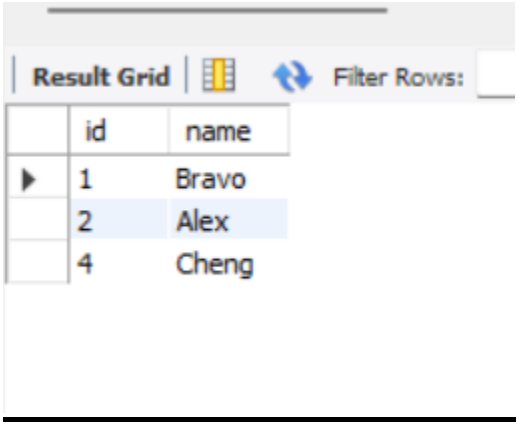
CLASS TABLE:

```
CREATE TABLE CLASS (
    ID INT,
    Name VARCHAR(50)
);
INSERT INTO CLASS VALUES (1, 'Bravo');
```

```
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');

SELECT * FROM CLASS;
```

SCREENSHOT:



Result Grid | Filter Rows:

	id	name
▶	1	Bravo
	2	Alex
	4	Cheng

EMPLOYEES Table

```
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
```

```
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct DECIMAL(4,2),
    Manager_Id INT,
    Department_Id INT
);

INSERT INTO EMPLOYEES VALUES
(101, 'Anaya', 'Singh', 'anaya@gmail.com', '9876543210', '2024-01-10', 'DEV', 50000, 0.10, 201, 10),
(102, 'Riya', 'Sharma', 'riya@gmail.com', '9876543211', '2023-05-15', 'HR', 40000, 0.05, 202, 20),
(103, 'Karan', 'Mehta', 'karan@gmail.com', '9876543212', '2022-03-20', 'DEV', 60000, 0.12, 201, 10);
SELECT * FROM EMPLOYEES;
```

SCREENSHOT:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Anaya	Singh	anaya@gmail.com	9876543210	2024-01-10	DEV	50000	0.10	201	10
102	Riya	Sharma	riya@gmail.com	9876543211	2023-05-15	HR	40000	0.05	202	20
103	Karan	Mehta	karan@gmail.com	9876543212	2022-03-20	DEV	60000	0.12	201	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Client_master Table

```
CREATE TABLE Client_master (
    Client_No INT PRIMARY KEY,
    Name VARCHAR(50),
    City VARCHAR(50),
    State VARCHAR(50),
    Pincode INT,
    Bal_Due INT
);

INSERT INTO Client_master VALUES
(1, 'Rahul', 'Delhi', 'Delhi', 110001, 5000),
(2, 'Priya', 'Mumbai', 'Maharashtra', 400001, 7000),
(3, 'Arjun', 'Bangalore', 'Karnataka', 560001, 3000);
SECLECT*FROM CLIENT_MASTER;
```

SCREENSHOT:

Result Grid | Filter Rows: | Edit: | Exp

Client_No	Name	City	State	Pincode	Bal_Due
1	Rahul	Delhi	Delhi	110001	5000
2	Priya	Mumbai	Maharashtra	400001	7000
3	Arjun	Bangalore	Karnataka	560001	3000
NULL	NULL	NULL	NULL	NULL	NULL

Bangalore

START TRANSACTION:

INSERT INTO CLASS VALUES(5,’DAVID’);

SELECT *FROM CLASS;

SCREENSHOT;

Result Grid | Filter Rows:

id	name
1	Bravo
2	Alex
4	Cheng
5	David

SAVEPOINT

SAVEPOINT A;

117 18:59:03 SAVEPOINT A 0 row(s) affected 0.000 sec

UPADTE CALSS TABLE;

ALTER TABLE CLASS

ADD PRIMARY KEY (ID);

UPDATE CLASS

SET Name = 'Alexander'

WHERE ID = 2;

SELECT*FROM CLASS;

SCREENSHOT;

Result Grid | Filter Rows:

id	name
1	Bravo
2	Alexander
4	Cheng
5	David
NULL	NULL

Rollback to SAVEPOINT

Rollback to a;

SELECT * FROM CLASS;

SCREENSHOT;

Result Grid | Filter Rows:

id	name
1	Bravo
2	Alex
4	Cheng
5	David

COMMIT;

SCREENSHOT;

137	19:09:35	COMMIT	0 row(s) affected	0.000 sec

EMPLOYEE TABLE

```
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct DECIMAL(4,2),
    Manager_Id INT,
```

```
    Department_Id INT
);
INSERT INTO EMPLOYEES VALUES
(101, 'Anaya', 'Singh', 'anaya@gmail.com', '9876543210', '2024-01-10', 'DEV', 50000, 0.10, 201, 10),
(102, 'Riya', 'Sharma', 'riya@gmail.com', '9876543211', '2023-05-15', 'HR', 40000, 0.05, 202, 20),
(103, 'Karan', 'Mehta', 'karan@gmail.com', '9876543212', '2022-03-20', 'DEV', 60000, 0.12, 201, 10);
SELECT * FROM EMPLOYEES;
```

SCREENSHOT;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

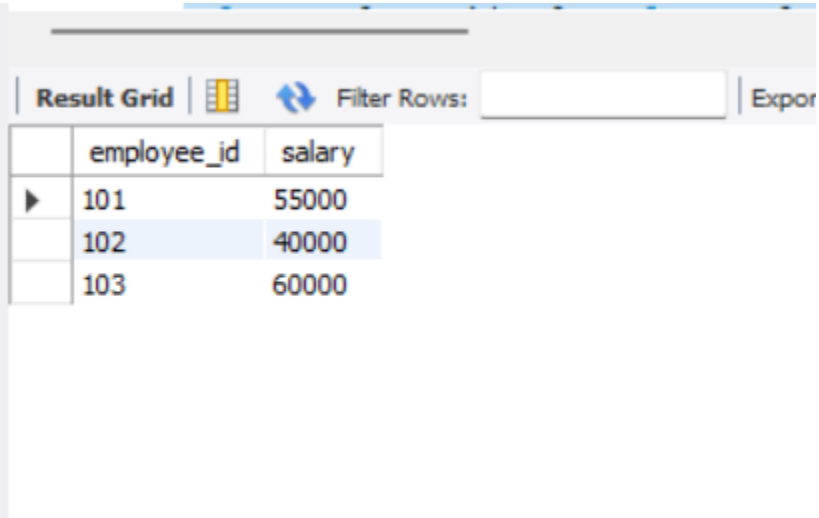
Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Anaya	Singh	anaya@gmail.com	9876543210	2024-01-10	DEV	50000	0.10	201	10
102	Riya	Sharma	riya@gmail.com	9876543211	2023-05-15	HR	40000	0.05	202	20
103	Karan	Mehta	karan@gmail.com	9876543212	2022-03-20	DEV	60000	0.12	201	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
START TRANSACTION;
SAVEPOINT emp_sp;
UPDATE EMPLOYEES
SET Salary = 55000
```

WHERE Employee_Id = 101;

select employee_id,salary from employees;

SCREENSHOT:



Result Grid Filter Rows: Expor

	employee_id	salary
▶	101	55000
	102	40000
	103	60000

ROLLBACK TO EMP_SP;

SELECT Employee_Id, Salary FROM EMPLOYEES;

SCREENSHOT;

Result Grid | Filter Rows:

Employee_Id	Salary
101	50000
102	40000
103	60000
NULL	NULL

COMMIT;

152 19:32:19 commit 0 row(s) affected 0.000 sec

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

mysqlexp.3*

```
create table class(
id int PRIMARY KEY,
name varchar(50)
);
```

Explanation:

- ID is defined as INTEGER and Primary Key.
- Name is defined as VARCHAR to store class name.

- PRIMARY KEY ensures unique identification of each record.

2. How do you create a table named EMPLOYEES with the specified attributes?

```sql
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct DECIMAL(4,2),
    Manager_Id INT,
    Department_Id INT
);
```

Explanation:

- Employee_Id is Primary Key.
- VARCHAR used for names and text fields.
- DATE used for Hire_Date.
- DECIMAL used for Commission_Pct.
- INT used for numeric values like Salary and Department_Id.

3. How do you create a table named Client_master with the specified attributes?

```sql
CREATE TABLE Client_master (
    Client_No INT PRIMARY KEY,
    Name VARCHAR(50),
    City VARCHAR(50),
    State VARCHAR(50),
    Pincode INT,
    Bal_Due INT
);
```

Explanation:

- Client_No is Primary Key.
- VARCHAR used for Name, City, State.
- INT used for numeric attributes like Pincode and Balance Due.

4. What SQL statement would you use to rename the table Client_master to Client12?

```sql
RENAME TABLE Client_master TO Client12;
```

Explanation:

- RENAME TABLE changes the table name permanently.
- After execution, Client_master will no longer exist.
- The table will now be accessed using Client12.

Show tables;

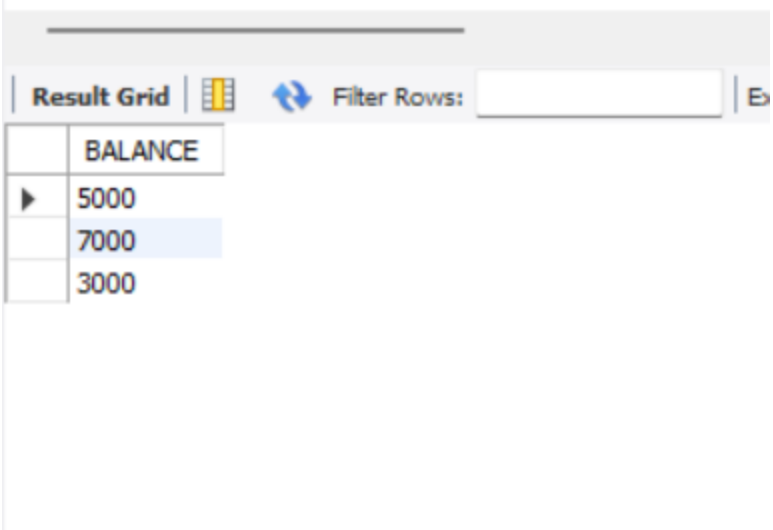
Result Grid | Filter Rows:

Tables_in_dbms_anshika
class
client12
department
employee
employees
iamarks
project
semsec
student
subject
works_on

5.How do you display the bal_due column heading as “BALANCE”?

SELECT bal_due AS BALANCE

FROM Client12;



Explanation:

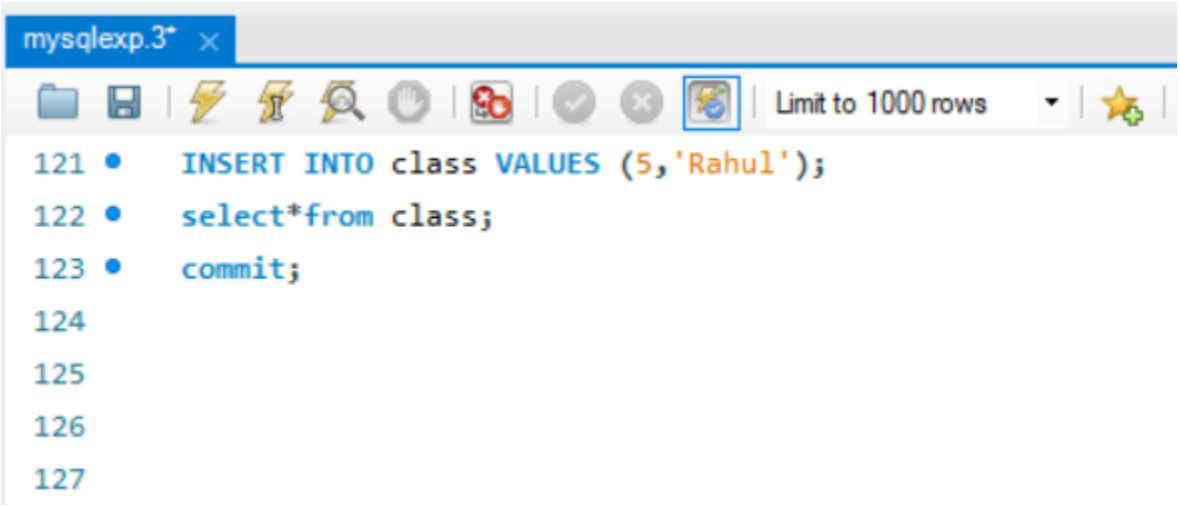
- AS keyword renames the column heading temporarily.
- It does NOT change the actual table structure.
- It only changes display output.

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;



Result Grid | Filter Rows:

id	name
1	Bravo
2	Alexander
4	Cheng
5	Rahul
NULL	NULL

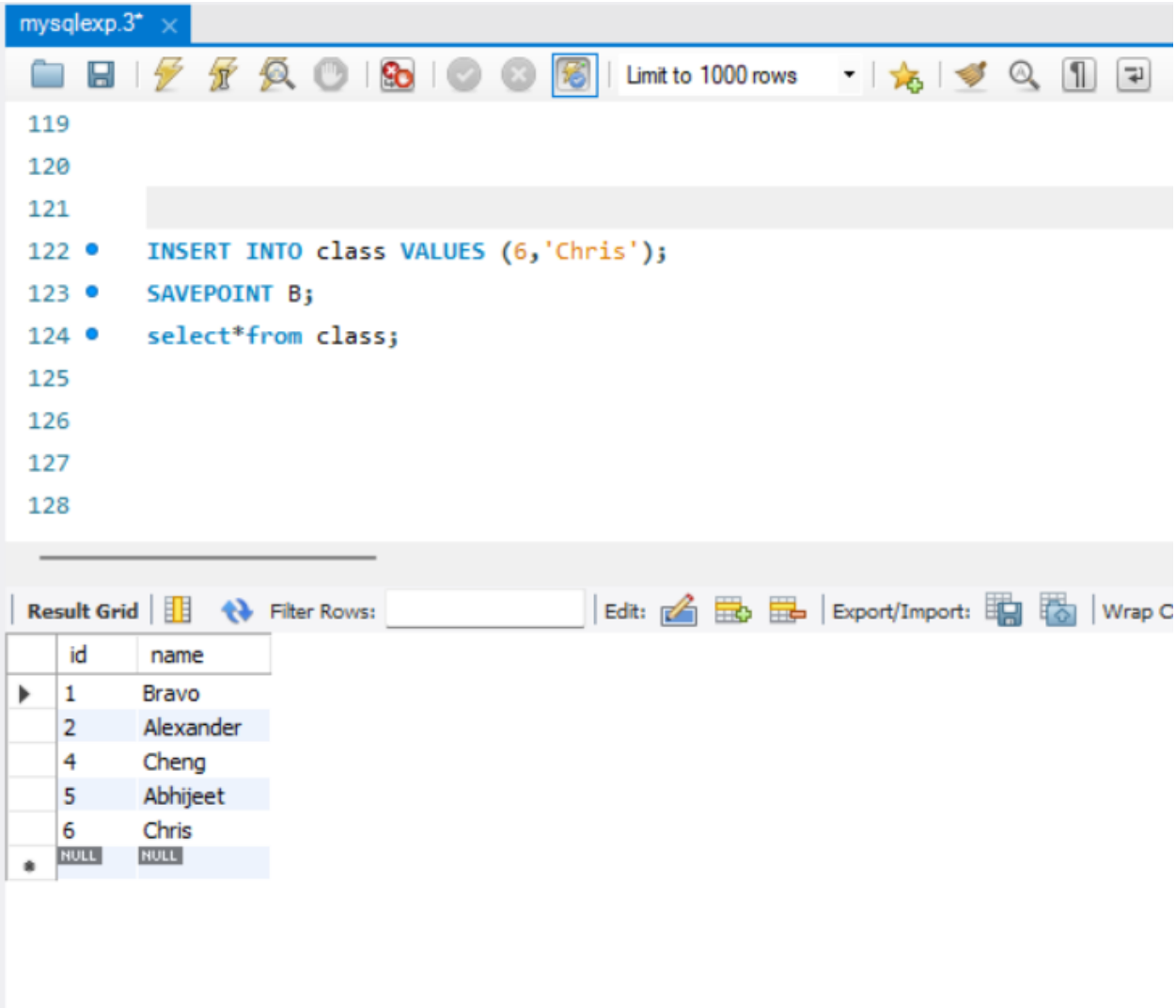
- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;
- SAVEPOINT A;

```
125
126 •   UPDATE class SET name = 'Abhijeet' WHERE id='5';
127 •   select*from class;
128 •   savepoint a;
129
130
```

Result Grid | Filter Rows: | Edit:

id	name
1	Bravo
2	Alexander
4	Cheng
5	Abhijeet
NULL	NULL

- INSERT INTO class VALUES (6,’Chris’);
- SAVEPOINT B;



- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;

168	20:09:43	INSERT INTO class VALUES (7,'bravo')	1 row(s) affected	0.000 sec
169	20:09:52	SAVEPOINT c	0 row(s) affected	0.000 sec

- SELECT * FROM class;

Result Grid | Filter Rows: | Edit: | Export/Import:

id	name
1	Bravo
2	Alexander
4	Cheng
5	Abhijeet
6	Chris
7	bravo
NULL	NULL

- ROLLBACK TO B;
- SELECT * FROM class;

```
rollback to b;
select*from class;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

id	name
1	Bravo
2	Alexander
4	Cheng
5	Abhijeet
6	Chris
NULL	NULL

- ROLLBACK TO A;

```
125 •  rollback to a;
126 •  select*from class;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

id	name
1	Bravo
2	Alexander
4	Cheng
5	Abhijeet
NULL	NULL

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

Result Grid | Filter Rows: | Edit: | Export/Impo

Employee_Id	First_Name	Last_Name	Salary
101	Anaya	Singh	50000
102	Riya	Sharma	40000
103	Karan	Mehta	60000
NULL	NULL	NULL	NULL

2. List out the employees who work under manager 100.

```
SELECT *FROM EMPLOYEES
WHERE Manager_Id = 100;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Find the names of the employees who have a salary greater than or equal to 4800.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
Anaya	Singh
Riya	Sharma
Karan	Mehta

4. List out the employees whose last name is ‘AUSTIN’.

```
SELECT *FROM EMPLOYEES
WHERE Last_Name = 'AUSTIN';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Find the names of the employees who work in departments 60, 70, and 80.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60, 70, 80);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
Anaya	Singh
Riya	Sharma
Karan	Mehta

6. Display the unique Manager_Id.

```
SELECT DISTINCT Manager_Id FROM EMPLOYEES;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Manager_Id
201
202

Explanation:

Displays unique manager IDs without repetition.

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

```
DESC CLIENT_MASTER;
INSERT INTO Client_master VALUES
(101, 'Rahul', 'Delhi', 'Delhi', 110001, 6000),
(102, 'Priya', 'Mumbai', 'Maharashtra', 400001, 4500),
(103, 'Arjun', 'Bangalore', 'Karnataka', 560001, 8000),
(104, 'Sneha', 'Chennai', 'Tamil Nadu', 600001, 3000),
(105, 'Karan', 'Hyderabad', 'Telangana', 500001, 7000);
SELECT*FROM CLIENT_MASTER;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Client_No	Name	City	State	Pincode	Bal_Due
1	Rahul	Delhi	Delhi	110001	5000
2	Priya	Mumbai	Maharashtra	400001	7000
3	Arjun	Bangalore	Karnataka	560001	3000
101	Rahul	Delhi	Delhi	110001	6000
102	Priya	Mumbai	Maharashtra	400001	4500
103	Arjun	Bangalore	Karnataka	560001	8000
104	Sneha	Chennai	Tamil Nadu	600001	3000
105	Karan	Hyderabad	Telangana	500001	7000
NULL	NULL	NULL	NULL	NULL	NULL

2. Find the names of clients whose bal_due > 5000.

```
SELECT Name
FROM Client_master
WHERE Bal_Due > 5000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Priya
Rahul
Arjun
Karan

3. Change the bal_due of ClientNO "C123" to Rs. 5100.

```
UPDATE Client_master
SET Bal_Due = 5100
WHERE Client_No = 'C123';
SELECT *FROM CLIENT_MASTER;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Client_No	Name	City	State	Pincode	Bal_Due
1	Rahul	Delhi	Delhi	110001	5000
101	Rahul	Delhi	Delhi	110001	5100
102	Priya	Mumbai	Maharashtra	400001	4500
103	Arjun	Bangalore	Karnataka	560001	8000
104	Sneha	Chennai	Tamil Nadu	600001	3000
105	Karan	Hyderabad	Telangana	500001	7000
2	Priya	Mumbai	Maharashtra	400001	7000
3	Arjun	Bangalore	Karnataka	560001	3000
C123	Rohan	Delhi	Delhi	110001	5100
NULL	NULL	NULL	NULL	NULL	NULL